BUTACU IONEL-CATALIN

Machine Learning Engineer | Desktop & Enterprise Software Engineer

linkedin.com/in/catalin-butacu
github.com/CatalinButacu



butacu.catalin@yahoo.com
+40 753 320 964

Summary

Software Engineer with 4 years of experience building desktop and enterprise software in C++ and Qt, with machine learning engineering experience gained through MSc study, certifications, and applied ML projects. Built computer vision, RAG, and agentic LLM applications covering model training, evaluation, inference, vector search, and edge deployment.

Professional Experience

Gemini CAD Systems — Iași, RO

Software Engineer — June 2023 – Present

- Delivered production C++/Qt UI, networking, and event-driven desktop features using signal/slot and model-view.
- Implemented multi-instance crash recovery with shared-memory IPC and a watchdog crash handler process.
- Integrated REST APIs and bidirectional cloud sync via Qt Network with JSON, auth, retry, and sync recovery.
- Refactored legacy C++ modules with smart pointers, clearer ownership, and unit-test-backed debt reduction.
- Fixed crashes, memory leaks, and regressions through root cause analysis, heap diagnostics, sanitizers, and tests.
- Collaborated in Agile Scrum for sprint planning, estimation, code review, demos, and retrospectives.

Software Engineer Intern — August 2022 – May 2023

- Built custom Qt UI components from specs and integrated them into production views.
- Extended real-time canvas behavior and optimized desktop event handling for smoother repaints.
- Delivered work through Git workflows, merge requests, and GitLab CI/CD for stable feature branches.

Special Telecommunications Service (STS) — Botoșani, RO

IT Technician – Electoral Bureau — August 2020 – September 2020

- Kept IT infrastructure operational across local electoral bureaus during the election process.
- Verified and transmitted electoral data through SIMPV and SICPV systems.
- Provided technical support and guidance to electoral bureau staff.

Education

Technical University of Iași — Iași, RO

Master of Science in Artificial Intelligence — October 2024 - June 2026

Advanced Coursework: *Machine Learning, Deep Learning, Natural Language Processing (NLP), Reinforcement Learning, Multi-Agent Systems, Cloud Computing, Big Data Techniques, Edge Computing Applications, Intelligent Cyber-Security*

Bachelor of Science in Systems Engineering — October 2020 - June 2024

Core Curriculum: *Advanced Computer Programming, Data Structures & Algorithms, Real-Time Programming, Database Management Systems, Data Processing, Microprocessor-Based Systems, Embedded Systems Engineering, Control Engineering, Computer Networks, Numerical Methods, Neural Networks, Optimization Techniques, Fuzzy Logic, Knowledge-Based Systems*

Independent Coursework: *DataTalksClub Zoomcamps in Machine Learning Engineering and Data Engineering — Docker, MLflow, FastAPI, Terraform, Kubernetes, dbt, Spark, Kafka, and Airflow.*

Certifications

- Oracle Cloud Infrastructure 2025 Data Science Professional (1Z0-1110-25)
- Oracle Cloud Infrastructure 2025 Generative AI Professional (1Z0-1127-25)
- Oracle Cloud Infrastructure 2025 AI Vector Search Professional (1Z0-184-25)
- Oracle Analytics Cloud 2025 Certified Professional (1Z0-1041-25)

Competencies

Software Engineering

- C++/Qt Desktop & Enterprise Development
- Software Architecture & Design Patterns
- Production Debugging & Root Cause Analysis
- Code Review, CI/CD & Agile Delivery
- Multithreading, Performance & System Design

Machine Learning & AI

- ML Model Training & Evaluation
- NLP, Generative AI & Prompt Design
- Deep Learning & Computer Vision
- LLM Applications, RAG & Vector Search
- Model Deployment & Inference Serving

Skills[1]

Languages: Python **4.0** ★ • C/C++ **4.5** ★ • SQL **3.5** ★ • Bash **3.5** ★ • TypeScript **3.0** ★

DataScience: PyTorch **4.0** ★ • Scikit-learn **3.5** ★ • NumPy **4.5** ★ • Pandas **4.0** ★ • OpenCV **3.0** ★

GenAI: HuggingFace **3.5** ★ • LangChain **3.5** ★ • LangGraph **3.5** ★ • Mesa **3.0** ★

DevOps: Docker **4.0** ★ • Linux **4.0** ★ • Git **4.0** ★ • GH Actions **3.5** ★ • Terraform **2.5** ★ • Kubernetes **2.0** ★

Backend: FastAPI **3.5** ★ • Pydantic **3.5** ★ • REST **3.5** ★ • gRPC **3.0** ★ • Node.js **3.0** ★ • pytest **2.0** ★

Pipelines: MLflow **3.5** ★ • Kafka **2.0** ★ • Airflow **2.0** ★ • Spark **2.0** ★ • dbt **2.0** ★

Desktop: Qt **4.5** ★ • MSBuild **4.0** ★ • STL **3.5** ★ • CMake **3.5** ★ • SonarQube **4.0** ★

Projects

Real-Time Generative Motion Modeling on SMPL-X ***(MSc Dissertation, in progress)*** **2026**
Real-time generative human motion modeling pipeline on SMPL-X with two core contributions: a Residual-FSQ tokenizer outperforming MoMask and a token-autoregressive S6/Mamba generator for bounded-memory streaming. Built as a modular Python training and inference stack with differentiable soft-decode losses, GloVe text encoding, and live SMPL-X visualization.

IDS/IPS Smart Firewall **2026**
Collaborative intelligent firewall system featuring a CNN threat classifier and an RL adversarial agent. Personal contribution focused on system architecture, Snort rule configuration, and developing a DBSCAN clustering module that analyzes runtime payloads to auto-generate rules, reducing ML inference load. Deployed on Ubuntu Server / Kali Linux.

Hand Gesture Recognition **2025**
YOLOv5n object-detection pipeline for hand gesture recognition (HaGRID dataset). TensorFlow workflow optimised for NPU/CPU deployment via TFLite INT8 quantisation, with CLAHE contrast preprocessing and temporal smoothing to reduce false positives. Developed and benchmarked on Android OS for NXP i.MX.

Med Assist — Romanian Pharmacy AI Assistant **2025–2026**
Full-stack RAG chatbot over the Romanian official ANMDM drug registry (7,555 entries). React + TypeScript frontend; **async** FastAPI + Pydantic service over Postgres. Hybrid retrieval — FAISS dense embeddings, BM25 sparse search, cross-encoder reranker — feeds Google Gemini. Designed prompt templates, evaluation checks and a tool-routing safety layer to ground recommendations in verified data. Auth0 JWT, Docker, Terraform-managed cloud, pytest with mocked LLM/DB layers.

Metaheuristic Framework **2025**
Python library implementing Canonical GA, Real-Valued GA, Differential Evolution, and Random Search. Each solver is benchmarked on standard mathematical test functions with automated CSV reporting and convergence visualisation.

fMRI Brain-Age Prediction **2025**
Competition ML pipeline predicting biological brain age from ∼20k fMRI neuroimaging connectivity features. Gender-stratified ElasticNet model with spectral feature selection; ranked 8th nationally at the WiDS Datathon 2025 University Edition.

Volunteer Experience

LOCAL BEST Group Iași Iași, RO
IT Department Coordinator Sep 2023 – Aug 2024

- Directed IT operations and maintained technical infrastructure across organisational events.
- Mentored volunteers and oversaw delivery of technical projects.
- Administered hardware, software licensing, and network resources.

Event Coordinator | JobShop Career Event Nov 2022 – Apr 2023

- Coordinated logistics and operations for a large student–industry career event.
- Managed scheduling, partner engagement, and on-site execution.
- Improved attendee experience through strategic planning and workflow optimisation.

TOPICS Responsible | Prove It 24H Techathon Dec 2021 – Mar 2022

- Defined technical requirements and coordinated hackathon logistics.
- Supported participants with rules, resources, and real-time assistance.
- Facilitated communication between organisers, industry mentors, and teams.

IT Responsible | BEST Training Weekend Jan 2021 – May 2022

- Developed and maintained the event website using HTML, CSS, and JavaScript.
- Collaborated with design and marketing responsible members on content, branding, and technical setup.

[1] *Scale (out of 5):* **5** owned in production **4** strong hands-on use **3** comfortable, still learning **2** coursework-level / guided exposure **1** foundational awareness